

복사전달 특강

Special Topics in Radiative Transfer

Lecture 10
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Based on: Chapter 8 of Noebauer & Sim (Living Reviews in Computational Astrophysics)

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Chapter 8: MCRT in Outflows and Explosions

Moving media — reference frames, Sobolev approximation, expansion work

- Previous chapters assumed static media or low-velocity environments

Many astrophysical systems involve **large material velocities**: stellar winds, SN ejecta, accretion disk winds

Special relativistic effects — especially the **Doppler effect** — become crucial

- Key challenge: frequency-dependent opacities must be evaluated in the correct frame

8.1 Mixed-Frame Approach

- Two frames: Lab Frame (LF) and Comoving Frame (CMF)
- Packet propagation in LF
- Interactions computed in CMF
- Doppler + aberration transforms

8.2 Line Interactions

- Sobolev approximation
- Optical depth integration simplified
- Re-emission direction sampling
- Test: H Lyman α profile

8.3 Expansion Work

- Energy exchange in moving media
- Packets gain/lose energy in LF
- MCRT naturally captures expansion work

Why Moving Media Require Special Treatment

Breakdown of the static-medium approximation

In **static media**: opacity χ is frequency-independent over the packet trajectory $\rightarrow$ simple integration

In **moving media**: CMF frequency ν_0 changes continuously as the packet propagates through a velocity gradient

- The optical depth integral (Eq. 79) must account for the Doppler-varying opacity along the path

$$\tau(l) = \int_{\nu_0(0)}^{\nu_0(l)} d\nu_0 \frac{dl}{d\nu_0} \frac{\nu_0}{\nu} \chi_0(\nu_0) \quad [\text{Eq. 79, Noebauer \& Sim}]$$

The factor ν/ν_0 transforms the opacity from CMF to LF; $d\nu_0/dl$ encodes the Doppler shift along the path.

General Case: Computationally Expensive

- Must integrate frequency change along trajectory
- Opacity varies strongly near line resonances
- Requires re-evaluation at each propagation step

Sobolev Approximation (Sect. 8.2)

- Velocity field monotonically varying $\rightarrow$ each line resonance encountered once
- Reduces to a purely local problem at the Sobolev point
- Enables dramatic computational speedup

8.1 The Mixed-Frame Approach

Lab Frame (LF) propagation + Comoving Frame (CMF) interactions

Lab Frame (LF): defined so the computational grid is at rest — natural for tracking positions and times

Comoving Frame (CMF): local rest frame of the material — interaction physics takes simplest form

- Mixed-frame: propagate packets in LF, but transform to CMF for each interaction

****Key idea**:** the handling of different tasks in MCRT simulations is easier in one or the other frame.

→ Spatial/temporal grid: defined in LF | Opacity/emissivity: evaluated in CMF

Doppler Shift (LF → CMF)

$$\nu_{0,i} = \gamma \nu_i (1 - \beta \cdot \mathbf{n}_i) \quad [\text{Eq. 71}]$$

- $\beta = v/c$ (material velocity)
- $\gamma = (1 - \beta^2)^{-1/2}$
- $\nu_{0,i}$: incident CMF frequency
- $\mathbf{n}_i$: incident propagation direction (LF)

Energy Transform (LF → CMF)

$$\varepsilon_{0,i} = \gamma \varepsilon_i (1 - \beta \cdot \mathbf{n}_i) \quad [\text{Eq. 72}]$$

- Coherent scattering in CMF: $\varepsilon_{0,e} = \varepsilon_{0,i}$
- Re-transform after interaction (Eqs. 75–76):

$$\varepsilon_e = \gamma \varepsilon_{0,e} (1 + \beta \cdot \mathbf{n}_{0,e})$$

$$\nu_e = \gamma \nu_{0,e} (1 + \beta \cdot \mathbf{n}_{0,e})$$

8.1 Aberration and Packet Initialization

Direction transforms and CMF initialization of packets

Aberration: propagation direction transforms between LF and CMF (in addition to Doppler)

- Must be taken into account when converting incident direction to CMF and emergent direction back to LF

Direction Transform LF → CMF [Eq. 77]

$$\mathbf{n}_{0,i} = \left(\frac{\nu}{\nu_0} \right) \left[\mathbf{n}_i - \gamma\beta \left(1 - \frac{1}{c} \frac{\gamma}{\gamma + 1} \mathbf{n}_i \cdot \mathbf{v} \right) \right]$$

- Used to convert direction before interaction
- Necessary for $O(v/c)$ accuracy

Direction Transform CMF → LF [Eq. 78]

$$\mathbf{n}_e = \left(\frac{\nu}{\nu_0} \right) \left[\mathbf{n}_{0,e} + \gamma\beta \left(1 + \frac{1}{c} \frac{\gamma}{\gamma + 1} \mathbf{n}_{0,e} \cdot \mathbf{v} \right) \right]$$

- Applied after interaction in CMF
- Re-emission direction set in CMF first

Optical Depth Accumulation (Moving Media)

- Packets propagate in LF accumulating τ
- Line opacities transformed: $\chi_{\text{LF}} = \chi_{\text{CMF}}(\nu_0/\nu)$
- CMF frequency changes continuously along path
- General case requires integrating Eq. (79)

CMF Packet Initialization

- Often natural to initialize packets in CMF
e.g., thermal emission sampled from Planck in CMF
- Then transform to LF: apply Eqs. 75–78 in reverse
- Ensures correct energy conservation from the start

8.2 Line Interactions in Outflows: Sobolev Approximation

Simplifying bound-bound opacities in monotonically expanding flows

Bound-bound (line) processes are **resonant**: only photons with CMF frequency $\approx \nu_{lu}$ interact

In a monotonically expanding flow, photons redshift continuously — each line resonance is encountered **once**

- Over the line width, the plasma state is approximately constant $\rightarrow$ Sobolev approximation

Sobolev approximation: replace the full frequency-dependent line profile by a delta-function at ν_{lu} .

$\rightarrow$ Optical depth integration collapses to a single local evaluation at the Sobolev point r_s .

Sobolev Optical Depth [Eq. 81]

$$\tau_s = \frac{\pi e^2}{m_e c} f_{lu} n_l \left(1 - \frac{n_u}{n_l} \frac{g_l}{g_u} \right) \left(\frac{dl}{d\nu_0} \right)_{r_s}$$

- f_{lu} : absorption oscillator strength $l \rightarrow u$
- n_l, n_u : level populations
- g_l, g_u : statistical weights
- $(dl/d\nu_0)$: velocity gradient term [Eq. 82]

Homologous Flow Simplification [Eq. 82]

$$\left(\frac{dl}{d\nu_0} \right)_{r_s} = \frac{c}{\nu_{lu}} \left[\frac{1}{(1 - \mu^2)v/r + \mu^2 dv/dr} \right]_{r_s}$$

- In homologous flow $v/r = \text{const}$
- Escape probability becomes direction-independent
- Greatly simplifies re-emission direction sampling

8.2 MCRT Implementation of the Sobolev Approximation

Packet propagation scheme with Sobolev line interactions

MC packets accumulate **continuum** optical depth τ_{cont} as they propagate (frequency-independent part)

Simultaneously, track distance to next **Sobolev point** for each line in a frequency-ordered list

- Decision at each step: does the packet reach a Sobolev point before τ_{cont} is exhausted?

Continuum Interaction (τ_{cont} reached first)

- Packet moved to distance l [Eq. 44]
- Interaction type sampled from continuum processes
- Sobolev line contributions reset/recalculated
- e.g., Thomson scattering, photoionization

Line Interaction (Sobolev point reached)

- Packet moved to Sobolev point r_s
- τ_{cont} instantaneously incremented by full τ_s
- Sample: scatter (resonant) or pass through
- Re-emission direction drawn from escape probability:

- $\rho(\mathbf{n}) \propto \frac{1 - \exp(-\tau_s(\mathbf{n}))}{\tau_s(\mathbf{n})}$ [Eq. 83]

The Sobolev approach stores lines in a frequency-ordered list (Abbott & Lucy 1985; Mazzali & Lucy 1993). As the packet redshifts monotonically, it sweeps through lines one by one — a hugely efficient scheme.

8.2 Sobolev Optical Depth Accumulation

Fig. 7 — Illustration of the optical depth accumulation process

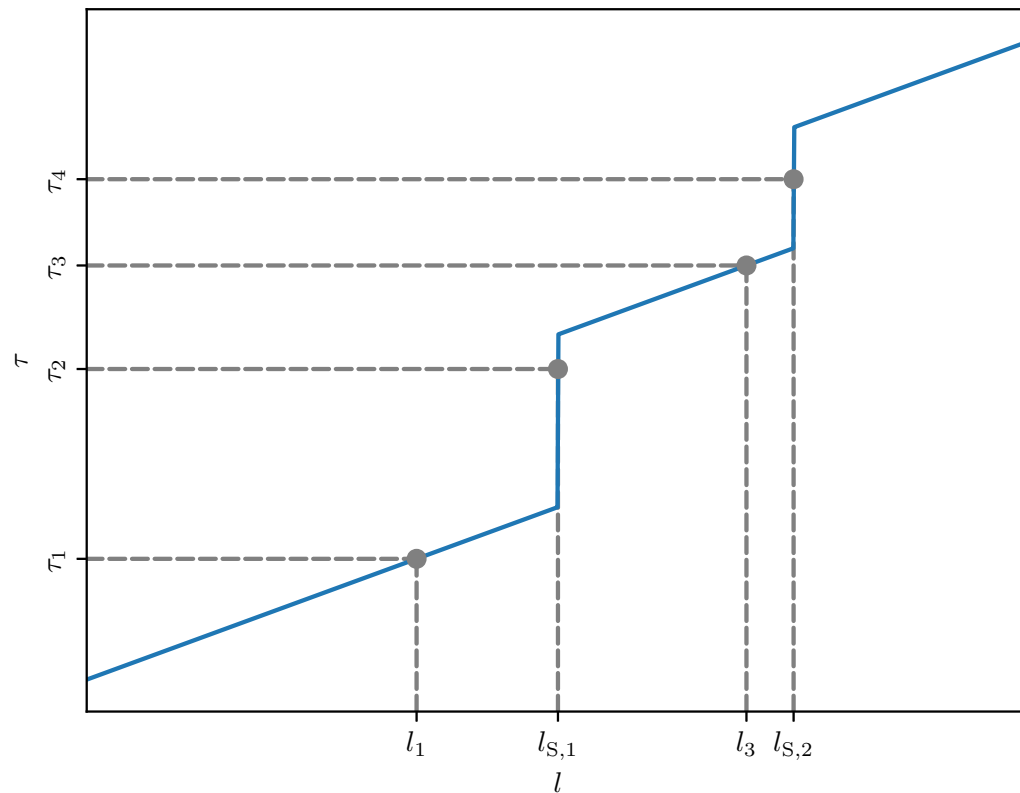


Fig. 7: Optical depth τ accumulates continuously from continuum processes. At two Sobolev points S_1 and S_2 , τ is instantaneously incremented by the full Sobolev line depth. For τ_1 and τ_3 : packet undergoes a continuum interaction at l_1 or l_3 . For τ_2 and τ_4 : packet undergoes a line interaction at the corresponding Sobolev point.

8.2 Test: H Lyman α Line Profile in a Homologous Flow

Fig. 8 — Validation of the Sobolev-based MCRT scheme

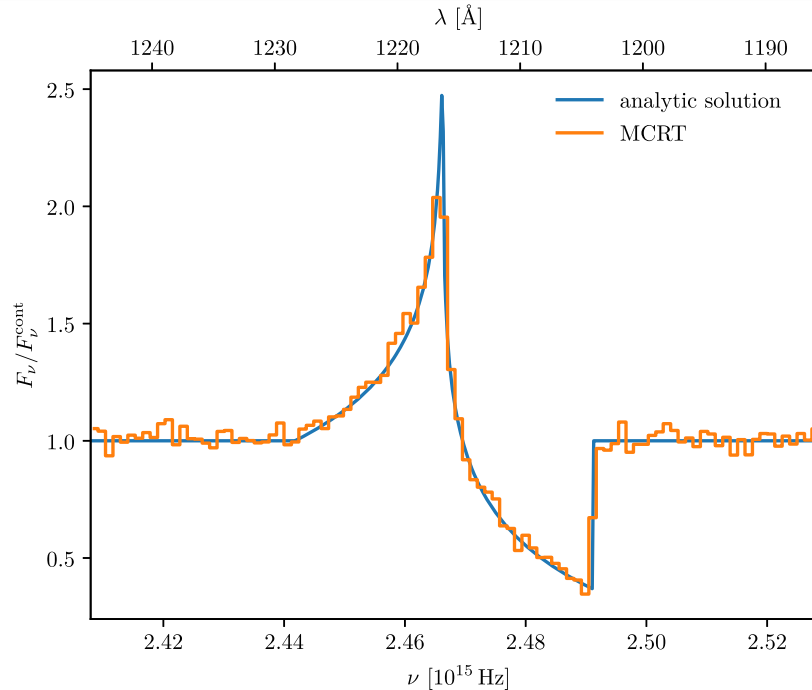


Fig. 8: H Lyman α line profile from a simple mixed-frame MCRT simulation. Blue: analytic solution (formal integration, Jeffery & Branch 1990). Orange: MCRT result. Emergent spectra normalized by the incident continuum flux F_ν^{cont} . Excellent agreement validates the mixed-frame + Sobolev MCRT approach.

Test Setup (Appendix A.2)

- H sphere, $\tau_s = 1$ for Ly α , $N = 10^5$ packets
- Analytic: Jeffery & Branch (1990)

Result

- MCRT matches analytic solution closely
- MC noise visible but statistically valid

8.3 MCRT and Expansion Work

Energy exchange in mass outflows: the radiation field drives the expansion

In RE, packet energy is conserved in the **CMF** during interactions

But in the **LF**, energy is NOT conserved: packets can gain or lose energy at each scattering

Photons propagating **outward** in an outflow lose energy on average → this is **expansion work**

Mean LF energy after isotropic re-emission from a packet with incident direction μ_i (Eq. 87):

$$\frac{\bar{\varepsilon}_e}{\varepsilon_i} = \frac{1}{2\beta} (1 - \mu_i\beta) [\log(1 + \beta) - \log(1 - \beta)] \rightarrow \frac{\bar{\varepsilon}_e}{\varepsilon_i} = 1 - \mu_i\beta + \mathcal{O}(\beta^2) \text{ for small } \beta$$

For $\mu_i = 1$ (outward propagating): $\bar{\varepsilon}_e/\varepsilon_i < 1$ — net energy loss → expansion work!

Physical Interpretation

- Radiation field initially aligned with outflow
- After multiple scatterings, packets lose energy in LF
- Energy transferred to kinetic energy of the outflowing gas
- This is the radiative driving mechanism in hot-star winds

MCRT Naturally Captures This

- No special treatment needed in mixed-frame MCRT
- Expansion work = $\sum (\varepsilon_i - \varepsilon_e)$ over all interactions
- Tracked by comparing incident and emergent LF energies
- Lucy (2005): model SN test demonstrates this directly

8.3 Test: Lucy (2005) Model Supernova

Figs. 9 & 10 — Energy flows and expansion work in a simplified SN ejecta model

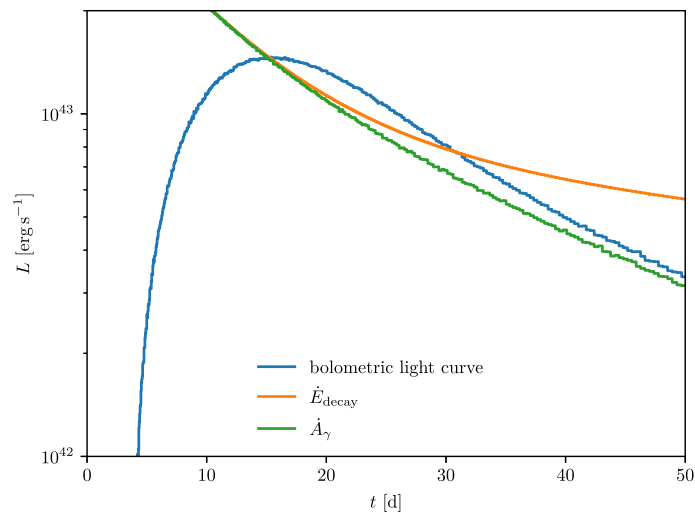


Fig. 9 (left): Bolometric light curve from the MCRH code for the Lucy (2005) model SN. Blue: escaping luminosity. Orange: energy release rate from ^{56}Ni radioactive decay chain. Green: rate at which γ -packets deposit energy into the UV-optical-IR radiation field.

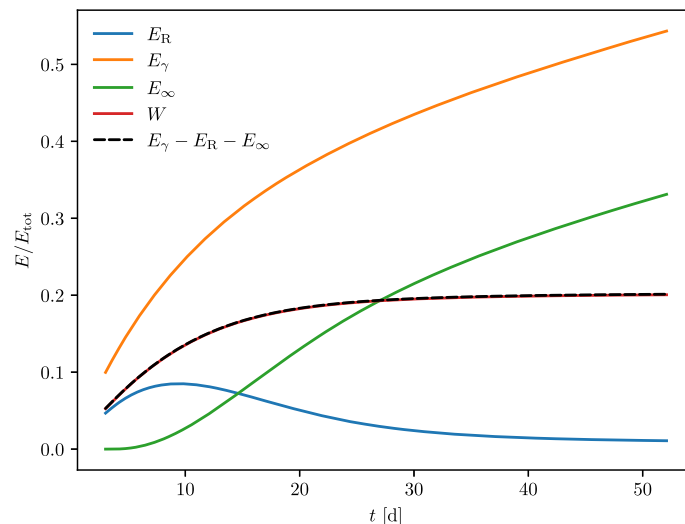


Fig. 10 (right): Energy flows — $E_R(t)$ (radiation field energy), $E_\infty(t)$ (escaped energy), $E_\gamma(t)$ (radioactive decay energy), $W(t)$ (expansion work term). The black dashed curve $E_\gamma - E_R - E_\infty$ precisely follows W , confirming energy conservation.

Key result: Mixed-frame MCRT correctly captures expansion work without special treatment.
Energy conservation verified at every time step across the simulation.

Chapter 8 — Summary

MCRT in outflows: mixed-frame, Sobolev, expansion work

Mixed-Frame Approach

- LF propagation + CMF interactions
- Doppler (Eqs. 71–76) + aberration (Eqs. 77–78)
- Optical depth accounts for CMF frequency drift
- Packet initialization often in CMF

Sobolev Approximation

- Monotonic velocity field → each line once
- τ_s : purely local evaluation at Sobolev point
- Line interactions: sampled via escape probability
- H Lyman α test: excellent agreement

Expansion Work

- LF energy NOT conserved per interaction
- Net energy loss = expansion work done on gas
- MCRT captures this automatically
- Validated with Lucy (2005) model SN

- These techniques are widely used: Abbott & Lucy (1985), Mazzali & Lucy (1993), Vink et al. (1999, 2000), Sim (2004), Noebauer & Sim (2015)...

MCRT + Sobolev is **the** standard approach for spectral synthesis in SN ejecta and hot-star winds

Next: how to **extract** physical information from MCRT simulations more efficiently (Chapter 9)